

Executive Summary

The following project is built for an end-to-end analytics solution to assess Annual Maintenance Contracts revenue and churn with the sales performance Atlas Heavy Machinery.

Business Problem

Atlas Heavy Machinery (AHM) sells construction equipment through 150 authorized dealers across West and Central India. The Annual Maintenance Contracts represent the company's only source of predictable recurring income. The Regional Sales Head has flagged a growing concern over AMC renewal rates are under pressure, and the team does not know whether churn is driven by the type of customer, or by the size of the fleet purchased, or the dealer managing the relationship. Additionally, there is no clarity on whether dealers who sell more machines are also generating proportionally more AMC revenue.

Objective

To investigate the drivers of AMC churn and the relationship between dealer sales volume and AMC revenue to help the Regional Sales Head make targeted retention decisions before the next quarterly business review.

Dataset Overview

Note: ER diagram for the schema has been uploaded to the repository.

4,000 machine sales records | 2,516 AMC contracts | Period: Apr-2023 to Mar-2024

Table	Description
Sales	Machine sale transactions
AMC	Annual Maintenance Contract records
Customers	Customer master with fleet and industry details
Dealers	Dealer master with region and status details
Machines	Machine master with category and pricing details

Assumptions & Limitations:

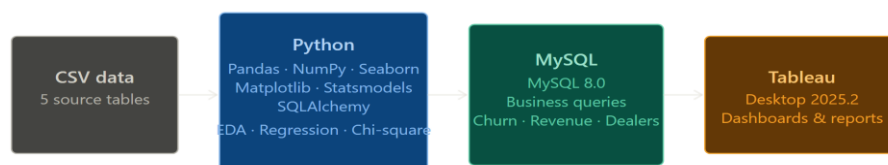
The data set contains sale records for 1 financial year starting from 1st April 2023 to 31st March 2024.

The data enables to analyse AMC.

The dataset was generated synthetically.

The name and logo of the organization do not represent any real entity. The scenario is an imitation of a real world scenario for portfolio building purposes.

Tech Stack Used:



Key Feature Engineering:

amc_signed = Y/N

Churn = Y/N

payment_mode = Cash/Financed

Treatment of Outliers and Null Values:

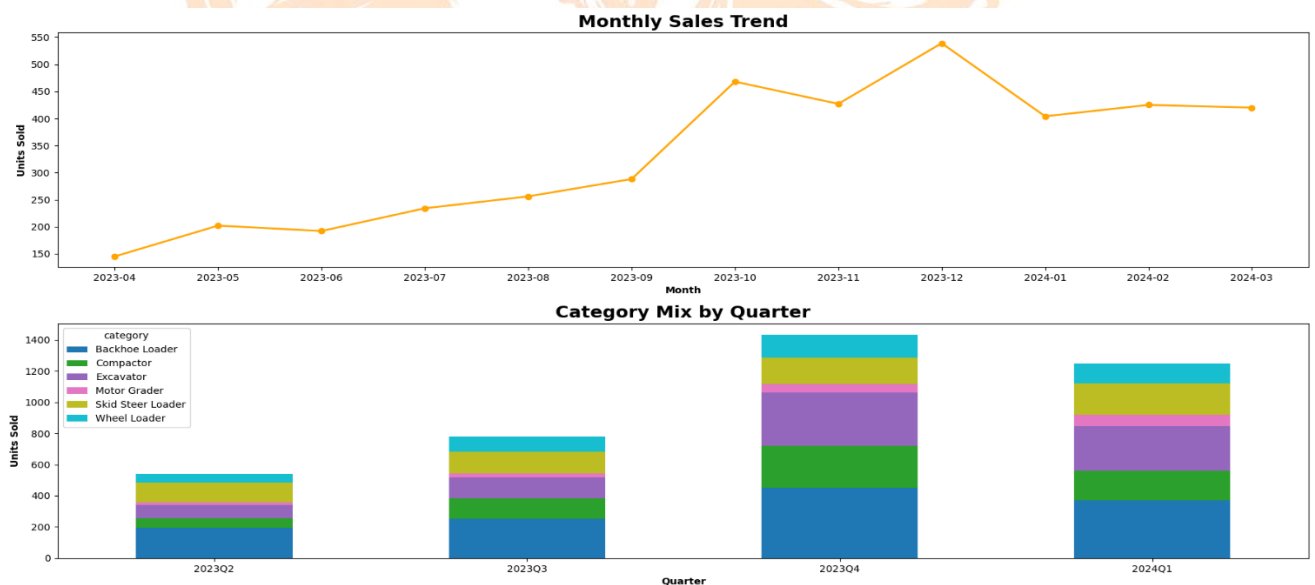
Note: Outliers were not treated as data issues, since extreme values represent real prices of the machines relevant to the analysis.

A total of 1030 records, which is 40.94%, have missing values in renewal_date column ; these records were kept to engineer churn flag and analyze the churn and so to prevent the loss of risk indicators. Similar step was followed for null values in the financier_name column which enabled identification of the cash purchases.

Analytics Findings From Exploratory Data Analysis:

category	Backhoe Loader	Compactor	Excavator	Motor Grader
Quarter				
2023Q2	191	64	85	18
2023Q3	253	131	133	27
2023Q4	451	270	339	55
2024Q1	372	187	287	72

category	Skid Steer Loader	Wheel Loader
Quarter		
2023Q2	124	57
2023Q3	139	95
2023Q4	171	148
2024Q1	201	130



- **Sales trend is upward**, peaking in December with **539 units**, showing strong year-end performance.
- **Backhoe Loader dominates every quarter**, making it the **top priority for AMC attachment**, while Excavator shows sharp growth from Q2 to Q4.
- **Compactor, Skid Steer, and Wheel Loader grow steadily**, while **Motor Grader stays low volume** all year.

category	total_sold	amc_signed	attachment_rate
Motor Grader	172	162	94.19
Wheel Loader	430	371	86.28
Excavator	844	708	83.89
Backhoe Loader	1267	746	58.88
Compactor	652	276	42.33
Skid Steer Loader	635	253	39.84

- **Motor Grader leads** with 94% AMC attachment, followed by Wheel Loader (86%) and Excavator (84%).
- **Backhoe Loader is the main concern** highest sales volume (1,267 units) but only close to 59% attachment rate.
- **Overall attachment is 63% approx**, with Compactor (42%) and Skid Steer (40%) lowest, showing need to improve AMC conversion.

Through Linear Regression we found:

Clarity on whether dealers who sell more machines are also generating proportionally more AMC revenue.

OLS Regression Results						
=====						
Dep. Variable:	amc_revenue_cr	R-squared:	0.552			
Model:	OLS	Adj. R-squared:	0.549			
Method:	Least Squares	F-statistic:	171.1			
Date:	Mon, 16 Mar 2026	Prob (F-statistic):	5.46e-26			
Time:	14:16:21	Log-Likelihood:	341.42			
No. Observations:	141	AIC:	-678.8			
Df Residuals:	139	BIC:	-672.9			
Df Model:	1					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	-0.0094	0.010	-0.967	0.335	-0.029	0.010
machines_sold	0.0044	0.000	13.080	0.000	0.004	0.005
=====						
Omnibus:	0.837	Durbin-Watson:	1.665			
Prob(Omnibus):	0.658	Jarque-Bera (JB):	0.873			
Skew:	-0.022	Prob(JB):	0.646			
Kurtosis:	2.617	Cond. No.	154.			
=====						

- Around **55% of AMC revenue depends on machine sales**, while the rest is driven by **dealer effort and conversion performance**.
- Each additional machine sold generates about **₹44,000 in AMC revenue**, confirming a strong business link between sales and AMC.
- Since **conversion rate is only 63%**, increasing sales alone is not enough **improving AMC attachment must be the key focus**.

Through Chi-Square test we found:

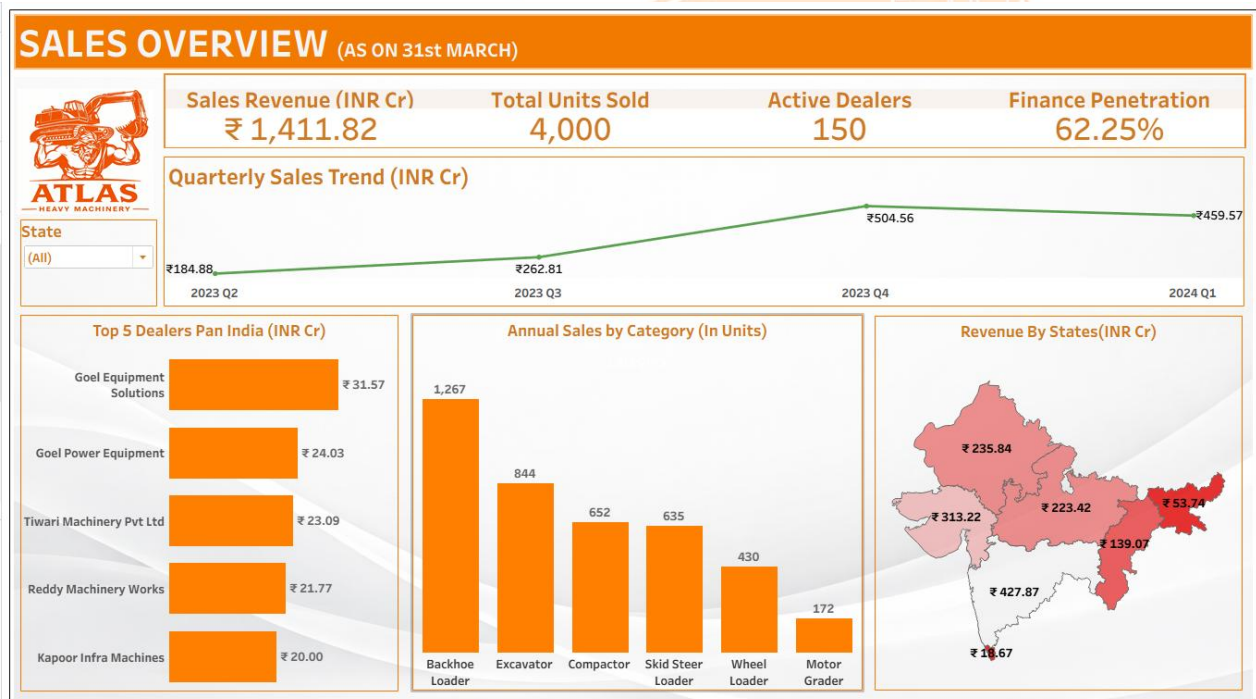
Clarity the impact of fleet size on customer churn.

churn	N	Y
fleet_size_category		
Large	387	32
Medium	475	114
Small	505	245
Very Large	119	2

TEST RESULTS
Chi-Square Statistic: 135.2855
P-Value: 0.0000
Degree of Freedom: 3.0000

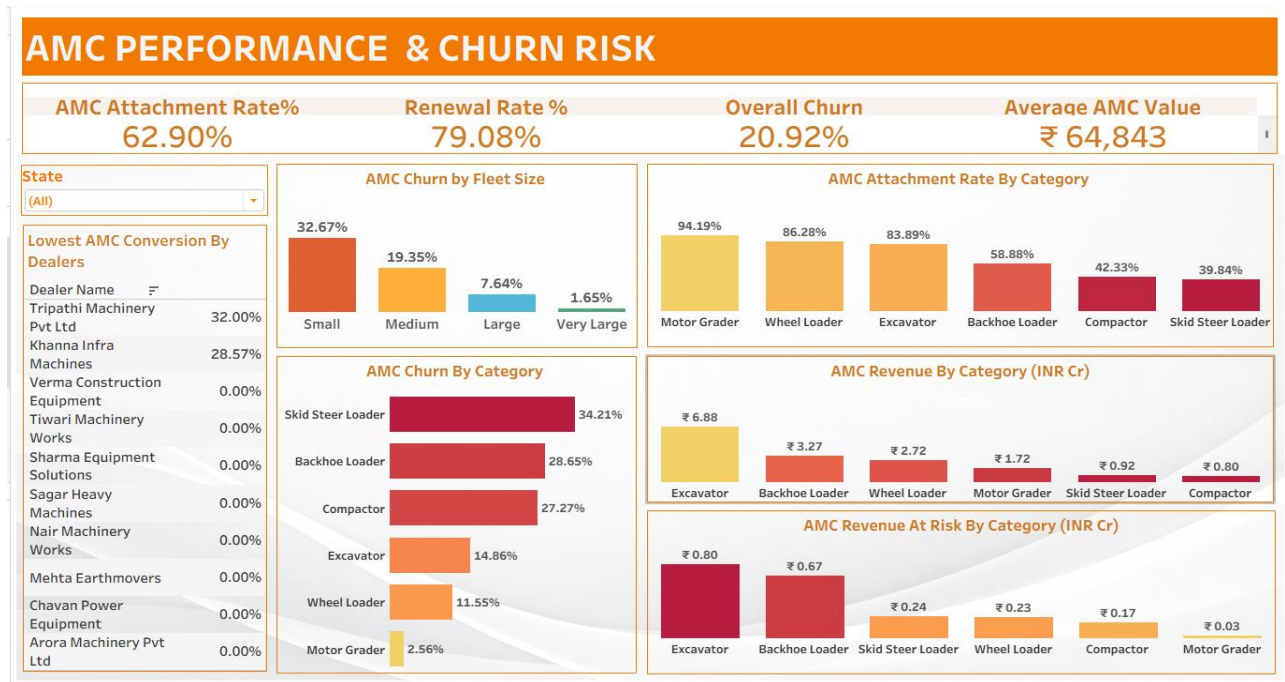
- **Fleet size strongly affects churn**, confirmed statistically ($p < 0.05$), meaning churn is not random across customers.
- **Small fleet customers churn the most**, while very large fleets have very low churn, showing clear risk concentration.
- **Fleet size can be used for retention strategy**, helping sales teams focus on high-risk small customers first.

Business Insights From Machine Sales Overview:



- The organization delivered **₹1,411.82 Cr revenue from 4,000 units sold through 150 dealers**, with **62.25% finance penetration**, and quarterly revenue peaked at **₹504.56 Cr in Q4**, exhibiting strong year-end momentum.
- **Backhoe Loader leads sales (1,267 units)**, dealer revenue is highest for **Goel Equipment Solutions (₹31.57 Cr)**, and performance is concentrated among a few key dealers.
- **State-wise revenue is led by Maharashtra (₹427.87 Cr), Gujarat (₹313.22 Cr), and MP (₹235.84 Cr)**, while East and South regions contribute lower revenue, indicating **regional concentration in sales performance**.

Business Insights From AMC Revenue & Churn Oversight:



- Overall **AMC attachment** is **62.90%** with **79.08%** renewal but **20.92%** churn, and average AMC value is **₹64,843**, indicating revenue loss despite good renewal, mainly due to low initial attachment and uneven dealer conversion.
- Churn is highest in Small fleets (32.67%)** and reduces with fleet size, dealer conversion varies significantly (lowest ~28–32%), and **Backhoe Loader, Compactor, and Skid Steer** show the **weakest attachment rates**, while **Motor Grader, Wheel Loader, and Excavator** have **strong conversion**, confirming both fleet size and category drive churn risk.
- Excavator generates highest AMC revenue (₹6.88 Cr)** and also highest revenue at risk (**₹0.80 Cr**), followed by Backhoe Loader (₹0.67 Cr), while **Skid Steer** shows highest churn %, meaning improving attachment and retention in high-volume and high-risk categories will have the biggest impact on AMC revenue.

HEAVY MACHINERY

Business Recommendations:

- Push AMC at point of sale:**
Make AMC bundling standard for **Backhoe Loader, Compactor, and Skid Steer**, and tie dealer incentives to **AMC attachment %**, not just sales volume in order to close conversion gaps.
- Focus on high-risk small fleets:**
Since **Small fleets have highest churn (32.67%)**, implement **early follow-ups, discounted renewals, and service engagement within first year** for customer retention.

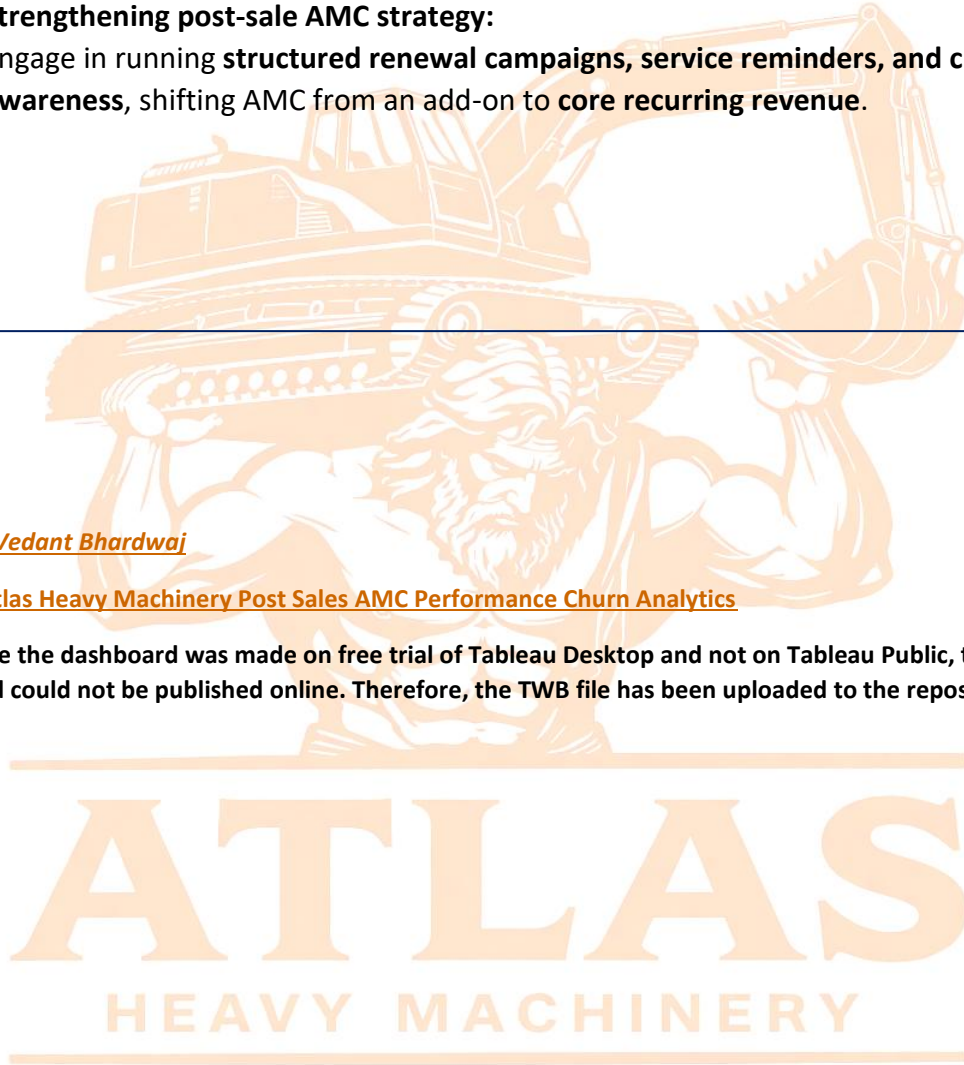
- **Fix dealer-level performance gaps:**
Introduction of **minimum AMC conversion benchmarks**, track dealer-wise performance, and take action on **low performers (28–32%)** through training or incentive restructuring.
- **Prioritize high revenue & high risk categories:**
Focus on **Excavator and Backhoe Loader**, as they contribute the most AMC revenue and risk, **small improvements here will drive optimum to maximum impact**.
- **Strengthening post-sale AMC strategy:**
Engage in running **structured renewal campaigns, service reminders, and customer awareness**, shifting AMC from an add-on to **core recurring revenue**.

Links:

LinkedIn: [Vedant Bhardwaj](#)

GitHub: [Atlas Heavy Machinery Post Sales AMC Performance Churn Analytics](#)

Note: Since the dashboard was made on free trial of Tableau Desktop and not on Tableau Public, the dashboard could not be published online. Therefore, the TWB file has been uploaded to the repository.



ATLAS
HEAVY MACHINERY